Tools — Code Intelligence

Symbol-aware search, references, and graph queries over the automatic codebase index (tree-sitter parsing, Tantivy FTS, semantic edge graph). All codeindex tools are read-only and hardwired always-allowed.

Tool	Description
`codeindex_search`	Search symbols by name/keyword.
`codeindex_symbols`	Query symbols with kind/file filters.
`codeindex_references`	Find all references to a symbol.
`codeindex_dependencies`	Query file import/dependent edges.
`codeindex_status`	Show index status and statistics.
`codeindex_reindex`	Trigger a full re-index.
`codeindex_explain`	Explain a graph node and its edges.
`codeindex_path`	Shortest path between two symbols.
`codeindex_communities`	Community detection over the graph.
`codeindex_godnodes`	Top-N most-connected symbols.

Use cases: finding function definitions, tracing callers, dependency analysis, code graph exploration.

System instruction: "MUST use codeindex instead of `grep` for code symbol queries. Use `grep` only for arbitrary text patterns."

Tools that need the semantic graph (`codeindex_explain`, `codeindex_path`, `codeindex_communities`, `codeindex_godnodes`) require the graph to be built first via the `/codeindex graph build` TUI command; they return a `codeindex_busy` response while the index is temporarily locked.

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codeindex_search

Search the codebase index for symbols, functions, types, and documentation.

Arguments

Argument	Type	Required	Description	Typical value
`query`	string	yes	Symbol name, keyword, or phrase to find	`"parse_config"`
`kind`	enum	no	Filter by symbol kind: `function`, `struct`, `enum`, `trait`, `impl`, `const`, `static`, `type_alias`, `module`, `macro`, `field`, `variant`, `interface`, `class`, `method`	`"function"`

Argument	Type	Required	Description	Typical value
`language`	string	no	Filter by programming language	`"rust"`, `"python"`, `"typescript"`
`file_pattern`	string	no	Filter by file path substring	`"src/parser"`, `".rs"`
`max_results`	integer	no	Maximum results (max 100)	`20`

Example:

```
codeindex_search query="parse_config" kind="function"
```

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codeindex_symbols

Query symbols (functions, structs, enums, traits) with location, signature, and documentation.

Arguments

Argument	Type	Required	Description	Typical value
`name`	string	no	Case-insensitive leading-fragment match	`"parse"`
`kind`	enum	no	Symbol kind (same list as `codeindex_search`)	`"struct"`
`file_path`	string	no	Filter by file path substring	`"crates/ragent-config"`
`language`	string	no	Filter by language	`"rust"`
`visibility`	enum	no	`public`, `private`, or `crate`	`"public"`
`limit`	integer	no	Maximum results (max 200)	`50`

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codeindex_references

Find all references to a symbol by name, grouped by file with reference kinds (`call`, `type`, `field_access`).

Arguments

Argument	Type	Required	Description	Typical value
`symbol`	string	yes	Symbol name to find references for	`"SessionState"`
`limit`	integer	no	Maximum references (max 200)	`50`

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codeindex_dependencies

Query file-level dependency edges from the index.

Arguments

Argument	Type	Required	Description	Typical value
`path`	string	yes	Relative file path	`"crates/ragent-agent/src/lib.rs"`
`direction`	enum	no	`imports` (what this file uses, default) or `dependents` (what uses this file)	`"dependents"`

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codeindex_status

Show whether the index is enabled, whether the FTS search index and semantic edge graph are built or building, plus files indexed, symbols extracted, languages, index size, and timestamps. No arguments required.

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codeindex_reindex

Trigger a full re-index of the codebase (scans all files, extracts symbols, updates the search index; may take a while on large repositories). Safe to call after bulk edits. No arguments required.

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codeindex_explain

Show a symbol's graph metadata (source file, line, community, degree) and up to 50 incoming/outgoing edges with kind and confidence tags.

Arguments

Argument	Type	Required	Description
`symbol`	string	yes	Name of the symbol to explain

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codeindex_path

Find the shortest path (by hop count) between two symbols, displayed as `A --kind--> B` hops with confidence tags.

Arguments

Argument	Type	Required	Description
`from`	string	yes	Source symbol name
`to`	string	yes	Target symbol name

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codeindex_communities

Run community detection over the code graph and display each community with its auto-generated label and member count. No arguments required.

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codeindex_godnodes

Show the top-N most-connected symbols (highest degree) with source files and edge counts.

Arguments

Argument	Type	Required	Description	Typical value
`n`	integer	no	Maximum god-nodes returned (max 100)	`10`

Example:

```
codeindex_search query="parse_config" kind="function"
codeindex_references symbol="parse_config"
codeindex_path from="parse_config" to="App::run"
```